

Yumeng He

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EDUCATION

University of Southern California

Master of Science

Aug 2024 – May 2026

Los Angeles, CA, USA

- Cumulative GPA: 3.83/4.0
- Computer Science major

University of Toronto

Honours Bachelor of Science with High Distinction

Sep 2019 – May 2024

Toronto, ON, Canada

- Cumulative GPA: 3.71/4.0
- Computer Science and Mathematics double major

RESEARCH EXPERIENCE

UCLA AIVC Lab Visiting Student | Full-time

June 2025 – present

Supervisor: Prof. Chenfanfu Jiang

Los Angeles, CA, USA

- Conducting full-time research on 3D articulated object generation, with a focus on neural representations of shape and joint dynamics for physically plausible synthesis
- Exploring physics-aware scene layout and hierarchical composition, combining structural priors with simulation constraints to guide object placement
- Preparing a first-author submission to CVPR 2026, presenting a unified framework for articulated object synthesis guided by both data-driven learning and physical consistency

USC RESL Lab Research Assistant | Part-time

July 2025 – present

Supervisor: Prof. Gaurav S. Sukhatme

Los Angeles, CA, USA

- Leading the Real2Sim pipeline for a robotic manipulation project, focusing on building a digital twin from a single RGB-D image of a cluttered tabletop scene—including object segmentation, mesh generation, and estimation of pose, scale, and placement—and importing the reconstructed scene into ManiSkill3 for simulation

WORK EXPERIENCE

Software Developer Intern | Full-time

Aug 2022 – Aug 2023

HCL Canada Inc.

Toronto, ON, Canada

- Automated build and deployment pipelines using Jenkins, accelerating production releases and minimizing manual errors by eliminating key bottlenecks in CI/CD
- Containerized microservices for both x86_64 (xLinux) and IBM PowerPC (pLinux) architectures, enabling architecture-agnostic builds and scalable, consistent deployments
- Integrated SonarQube into the pipeline; documented a detailed usage guide, presented to cross-functional teams, and resolved 3 critical bugs and 308 code smells (including 4 blockers, 41 major issues)
- Deployed a shell script to auto-clean legacy logs weekly (Sunday 00:00 EST), reducing disk clutter and improving system performance with zero manual intervention

PROFESSIONAL SERVICES

- Mentor for Viterbi Graduate Mentorship Program (Fall 2025)

AWARDS & HONORS

- Dean's List Scholar for 4 years (2020 - 2024)
- Ranked in the top 25% of contestants in the Galois Contest (2017)
- Ranked in the top 25% of contestants in the Cayley Contest (2017)
- Ranked in the top 25% of contestants in the Canadian Intermediate Mathematics (2016)

TECHNICAL SKILLS

- **Operating Systems:** Linux, Windows, MacOS
- **Programming Languages:** C++, Python, JavaScript, HTML, CSS, MySQL
- **Graphics and Simulation:** OpenGL, GLSL, Eigen, PyTorch
- **Tools and DevOps:** Docker, Jenkins, Gradle, Ant, Shell, Git, Github Actions
- **Languages:** Mandarin (native), English (fluent)